

Cost Sheet — Simple Guide

What is a Cost Sheet?

Think of a cost sheet as a **recipe for figuring out how much it cost to make and sell something**, built up in layers. You start with raw materials and keep adding costs in stages until you reach the final selling price.

The chain looks like this:

```
Raw Material Used
  → Prime Cost
  → Gross Factory Cost
  → Factory Cost
  → Cost of Production
  → Cost of Goods Sold
  → Cost of Sales
  → Sales (selling price)
```

Each arrow means "add some more costs and give the running total a new name."

1. Types of Costs (the vocabulary)

A) Direct versus Indirect

- **Direct cost** — you can point to it inside the finished product. Example: the steel in a car, the flour in bread.
- **Indirect cost** — it helps make the product, but you cannot trace it to one specific unit. Example: cleaning supplies, the factory supervisor's salary. Indirect costs get grouped together and called **overheads**.

Quick tests to decide direct versus indirect:

Test	Ask yourself...
Can I see it?	Is this material physically inside the finished product?
Is it worth measuring?	Even if I could trace it, is it practical to bother?
Does it matter?	Is the amount big enough to matter?

Test	Ask yourself...
Does it scale?	Does usage go up and down with how much I produce?

B) By Function — this is the part that actually builds the cost sheet

Category	Ask yourself...	Examples
Factory cost	Would this cost disappear if the factory shut down?	Direct material, direct labour, factory rent, factory electricity, machine depreciation
Administrative cost	Is this just "running the business" overhead?	Office salaries, audit fees, office rent
Selling cost	Is this about getting customers interested?	Advertising, sales staff salary and commission
Distribution cost	Is this about physically getting the product to the customer?	Freight, packing, warehousing
Research and Development cost	Is this about inventing something new?	Scientist salaries, lab materials, prototypes

C) By Behaviour (how cost changes with production)

Type	Total amount	Amount per unit
Fixed	Stays the same no matter how much you make	Goes down as you make more
Variable	Goes up and down directly with how much you make	Stays the same per unit
Semi-variable	Part fixed, part variable	Partly stays the same, partly changes

D) Controllable versus Uncontrollable

- **Controllable** — a manager can influence it. Example: overtime hours a supervisor allows.
- **Uncontrollable** — out of that manager's hands. Example: factory rent, government taxes.

E) Escapable versus Inescapable (only matters for decisions, not for the sheet itself)

- **Escapable** — goes away if you stop doing the activity.

- **Inescapable** — keeps happening regardless.

2. The Cost Sheet — Full Template

Every line is either **add** or **subtract**. Nothing else.

```
    Opening Stock of Raw Material
+   Purchases of Raw Material
+   Carriage Inwards (freight to bring material into the factory)
+   Import Duty
-   Closing Stock of Raw Material
-   Purchase Returns
-   Sale of Scrap
```

= RAW MATERIAL CONSUMED

```
+   Direct Labour
+   Direct Expenses (special plant hire, patents, design costs)
```

= PRIME COST

```
+   Factory Overheads (rent, depreciation, lighting, insurance - all factory-
related)
```

= GROSS FACTORY COST

```
+   Opening Stock of Work-in-Progress
-   Closing Stock of Work-in-Progress
```

= FACTORY COST

```
+   Administrative Overheads (office salaries, audit fees, director's fees)
```

= COST OF PRODUCTION

```
+   Opening Stock of Finished Goods
-   Closing Stock of Finished Goods
```

= COST OF GOODS SOLD

```
+   Selling and Distribution Overheads (advertising, sales commission, bad
debts, carriage outwards)
```

= COST OF SALES

+ Profit

= SALES

Two rules to remember:

1. Only **bad debts actually written off** count as a selling expense. A provision for doubtful debts, or debts you later recovered, do not count.
2. If you are only given a stock figure **in units** (not in rupees), you have to convert it — see the formulas below.

3. Formulas — Plain List

What you want	Formula
Raw Material Consumed	Opening Stock of Raw Material + Purchases + Carriage Inwards + Import Duty – Closing Stock of Raw Material – Purchase Returns – Scrap Sold
Prime Cost	Raw Material Consumed + Direct Labour + Direct Expenses
Gross Factory Cost	Prime Cost + Factory Overheads
Factory Cost	Gross Factory Cost + Opening Stock of Work-in-Progress – Closing Stock of Work-in-Progress
Cost of Production	Factory Cost + Administrative Overheads
Cost of Goods Sold	Cost of Production + Opening Stock of Finished Goods – Closing Stock of Finished Goods
Cost of Sales	Cost of Goods Sold + Selling and Distribution Overheads
Sales	Cost of Sales + Profit
Closing Stock of Work-in-Progress, in rupees (when only units are given)	$(\text{Gross Factory Cost} \div \text{Units Produced}) \times \text{Closing Stock of Work-in-Progress in units}$
Closing Stock of Finished Goods, in rupees (when only units are given)	$(\text{Cost of Production} \div \text{Units Produced}) \times \text{Closing Stock of Finished Goods in units}$
Closing Stock of Finished Goods, in units	Opening Stock of Finished Goods in units + Units Produced – Units Sold

Why the two conversion formulas work: once you know Gross Factory Cost or Cost of Production, dividing it by units produced gives you a "cost per unit." Multiply that by however many units are still sitting in stock, and you get the rupee value of that stock.

4. How to Solve Any Cost Sheet Question — Checklist

1. Go through the question and label every item: raw material, direct labour, direct expense, factory overhead, administrative overhead, selling or distribution overhead, a stock figure, a unit figure, or the sales figure.
 2. Draw the full template from Section 2 — every line, even if some are zero.
 3. Work out Raw Material Consumed, then Prime Cost.
 4. Add Factory Overheads to get Gross Factory Cost.
 5. Adjust for Work-in-Progress stock (convert from units if needed) to get Factory Cost.
 6. Add Administrative Overheads to get Cost of Production.
 7. Adjust for Finished Goods stock (convert from units if needed) to get Cost of Goods Sold.
 8. Add Selling and Distribution Overheads to get Cost of Sales.
 9. Add Profit to get Sales — or, if Sales is given instead, subtract Cost of Sales from it to get Profit.
-

5. Worked Example (numbers from the book)

Item	Amount (rupees)
Opening Stock of Raw Material	25,000
Purchases	48,000
Carriage Inwards	1,500
Import Duty	2,500
Closing Stock of Raw Material	15,000
Purchase Returns	1,000
Scrap Sold	500
Direct Labour	20,000
Direct Expenses (5,000 + 2,500 + 1,800 + 1,700)	11,000
Factory Overheads (2,500 + 3,500 + 1,700 + 1,850)	9,550
Opening Stock of Work-in-Progress	20,000

Item	Amount (rupees)
Closing Stock of Work-in-Progress	18,000
Administrative Overheads (3,750 + 1,800 + 2,500 + 1,850)	9,900
Opening Stock of Finished Goods	2,000
Closing Stock of Finished Goods	1,500
Selling and Distribution Overheads (11,000 + 3,500 + 2,700 + 1,850)	19,050
Sales	200,000

Build-up:

Stage	Calculation	Result (rupees)
Raw Material Consumed	$25,000 + 48,000 + 1,500 + 2,500 - 15,000 - 1,000 - 500$	60,500
Prime Cost	$60,500 + 20,000 + 11,000$	91,500
Gross Factory Cost	$91,500 + 9,550$	101,050
Factory Cost	$101,050 + 20,000 - 18,000$	103,050
Cost of Production	$103,050 + 9,900$	112,950
Cost of Goods Sold	$112,950 + 2,000 - 1,500$	113,450
Cost of Sales	$113,450 + 19,050$	132,500
Profit	$200,000 - 132,500$	67,500
Sales	given	200,000

6. Your Two Practice Questions — Full Verbose Walkthrough

Question 1

The data, laid out exactly as given:

Item	Value
Units Produced	15,000 units
Prime Cost	₹2,30,000
Factory Lighting, Rent and Rates	₹15,000

Item	Value
Depreciation on Factory Building	₹5,000
Insurance of Factory Building	₹2,500
Opening Stock of Work-in-Progress	₹1,500
Closing Stock of Work-in-Progress	1,000 units (rupee value NOT given)

What is actually being asked: the question wants two things — (a) Gross Factory Cost, and (b) Factory Cost. Nothing beyond that. So even though the full template in Section 2 has many more stages (Cost of Production, Cost of Goods Sold, Cost of Sales, Sales), we only need to travel as far as Factory Cost and then stop.

Step 1 — Label every item (per the Section 4 checklist, Step 1).

Going line by line: Units Produced is a unit figure, used later for the per-unit conversion. Prime Cost is already a finished sub-total, handed to us directly, so we do not need to build it up from Raw Material Consumed, Direct Labour, and Direct Expenses — the question has done that part for us. The three items — Factory Lighting/Rent/Rates, Depreciation on Factory Building, and Insurance of Factory Building — are all Factory Overheads, because they are all costs tied to running the factory building and would disappear if the factory shut down (this is exactly the test given in Section 1, Part B). Opening Stock of Work-in-Progress and Closing Stock of Work-in-Progress are the two Work-in-Progress adjustment lines that sit between Gross Factory Cost and Factory Cost in the template.

Step 2 — Since Prime Cost is already given, we start the template from that point onward (Section 2's template, picking up mid-way):

```

PRIME COST
+ Factory Overheads
-----
= GROSS FACTORY COST

+ Opening Stock of Work-in-Progress
- Closing Stock of Work-in-Progress
-----
= FACTORY COST

```

Step 3 — Add up the Factory Overheads.

This is a simple addition of the three factory-related items identified in Step 1:

Factory Overheads = Factory Lighting, Rent and Rates + Depreciation on Factory Building + Insurance of Factory Building
Factory Overheads = 15,000 + 5,000 + 2,500
Factory Overheads = **₹22,500**

Step 4 — Calculate Gross Factory Cost.

The formula, taken directly from the Section 3 formula table, is:

Gross Factory Cost = Prime Cost + Factory Overheads

Substituting the numbers:

Gross Factory Cost = 2,30,000 + 22,500
Gross Factory Cost = **₹2,52,500**

This is **Answer (a)**.

Step 5 — Deal with the Closing Stock of Work-in-Progress problem.

Here is the obstacle: the template's next step needs us to subtract the Closing Stock of Work-in-Progress in rupees, but the question has only given us the Closing Stock of Work-in-Progress in units — specifically, 1,000 units. We are not told what those 1,000 units are worth in rupees directly.

This is precisely the situation Section 3 anticipates with its special conversion formula:

Closing Stock of Work-in-Progress, in rupees = (Gross Factory Cost ÷ Units Produced) × Closing Stock of Work-in-Progress in units

The logic behind this formula (explained in Section 3's closing note) is this: Gross Factory Cost represents the total cost of everything produced during the period, spread across all 15,000 units produced. So if we divide Gross Factory Cost by the number of units produced, we get an average "cost per unit" figure. Once we know what one unit costs on average, we can multiply that per-unit cost by however many units are sitting unfinished in the Work-in-Progress stock at the end of the period, and that gives us the rupee value of that stock.

Sub-step 5a — Find the cost per unit.

Cost per unit = Gross Factory Cost ÷ Units Produced
Cost per unit = 2,52,500 ÷ 15,000
Cost per unit = 16.8333... (repeating)

We can round this to ₹16.83 per unit for display purposes, but it is important to carry the more precise, unrounded figure into the next multiplication step to avoid compounding rounding errors. The exact fraction is $252500/15000 = 16.83333...$

Sub-step 5b — Multiply by the number of units in the Closing Stock of Work-in-Progress.

Closing Stock of Work-in-Progress in rupees = $16.8333... \times 1,000$
Closing Stock of Work-in-Progress in rupees = **₹16,833.33** (rounded to the nearest paisa)

Step 6 — Calculate Factory Cost.

The formula from Section 3 is:

Factory Cost = Gross Factory Cost + Opening Stock of Work-in-Progress – Closing Stock of Work-in-Progress

Substituting every number we now have:

Factory Cost = $2,52,500 + 1,500 - 16,833.33$

Breaking this arithmetic into two smaller steps for clarity:

First, add Gross Factory Cost and Opening Stock of Work-in-Progress: $2,52,500 + 1,500 = 2,54,000$

Then subtract the Closing Stock of Work-in-Progress figure we just calculated: $2,54,000 - 16,833.33 = \mathbf{₹2,37,166.67}$

This is **Answer (b)**.

Final Answers for Question 1:

Answer	Value
(a) Gross Factory Cost	₹2,52,500.00
(b) Factory Cost	₹2,37,166.67

Question 2

The data, laid out exactly as given:

Item	Value
Units Produced	32,000 units
Units Sold	38,000 units
Selling Rate	₹15 per unit
Prime Cost	₹2,32,000
Factory Lighting, Rent and Rates	₹1,11,000
Depreciation on Plant and Machinery	₹27,500
Factory Lighting and Power	₹22,500
Opening Stock of Work-in-Progress	₹79,000
Closing Stock of Work-in-Progress	2,000 units (rupee value NOT given)
Director's Fees	₹13,000
Office Salaries	₹7,500
Audit Fees	₹2,500
Opening Stock of Finished Goods	3,500 units, valued at ₹35,000
Closing Stock of Finished Goods	NOT given at all — neither in units nor in rupees

What is being asked: this question does not explicitly limit itself to two sub-answers the way Question 1 did — it gives us enough information to walk much further down the template, potentially all the way to Sales and Profit. So we will proceed stage by stage, exactly as the Section 4 checklist prescribes, and go as far as the data honestly allows.

Step 1 — Label every item.

Prime Cost is again given directly, so no need to build it from Raw Material Consumed. Factory Lighting/Rent/Rates, Depreciation on Plant and Machinery, and Factory Lighting and Power are all Factory Overheads — each one is a cost of running the factory itself, satisfying the "would this disappear if the factory shut down?" test from Section 1, Part B. Opening and Closing Stock of Work-in-Progress are the usual adjustment pair between Gross Factory Cost and Factory Cost. Director's Fees, Office Salaries, and Audit Fees are all Administrative Overheads — none of them relate to making the product or selling it; they are the cost of running the business as a governing entity, which is exactly the test given in Section 1, Part B for Administrative Cost.

Opening and Closing Stock of Finished Goods are the adjustment pair between Cost of Production and Cost of Goods Sold. Units Sold and Selling Rate per unit will eventually let us calculate Sales, once we reach that stage.

Step 2 — Draw out the relevant portion of the template (Section 2), picking up from Prime Cost onward, since Prime Cost is already given:

```
PRIME COST
+ Factory Overheads
-----
= GROSS FACTORY COST

+ Opening Stock of Work-in-Progress
- Closing Stock of Work-in-Progress
-----
= FACTORY COST

+ Administrative Overheads
-----
= COST OF PRODUCTION

+ Opening Stock of Finished Goods
- Closing Stock of Finished Goods
-----
= COST OF GOODS SOLD
```

Step 3 — Add up the Factory Overheads.

Factory Overheads = Factory Lighting, Rent and Rates + Depreciation on Plant and Machinery
+ Factory Lighting and Power Factory Overheads = 1,11,000 + 27,500 + 22,500

Adding these in two smaller steps: 1,11,000 + 27,500 = 1,38,500, and then 1,38,500 + 22,500 = **₹1,61,000**

Step 4 — Calculate Gross Factory Cost.

Gross Factory Cost = Prime Cost + Factory Overheads

Gross Factory Cost = 2,32,000 + 1,61,000 = **₹3,93,000**

Step 5 — Convert the Closing Stock of Work-in-Progress from units into rupees.

Just like in Question 1, we are told the Closing Stock of Work-in-Progress only in units (2,000 units), not in rupees. We again reach for the Section 3 conversion formula:

Closing Stock of Work-in-Progress, in rupees = (Gross Factory Cost ÷ Units Produced) × Closing Stock of Work-in-Progress in units

Sub-step 5a — Cost per unit at the Gross Factory Cost stage:

Cost per unit = Gross Factory Cost ÷ Units Produced
Cost per unit = 3,93,000 ÷ 32,000
Cost per unit = 12.28125 (this is an exact decimal, not repeating)

Sub-step 5b — Multiply by the units in Closing Stock of Work-in-Progress:

Closing Stock of Work-in-Progress in rupees = 12.28125 × 2,000
Closing Stock of Work-in-Progress in rupees = **₹24,562.50**

Step 6 — Calculate Factory Cost.

Factory Cost = Gross Factory Cost + Opening Stock of Work-in-Progress – Closing Stock of Work-in-Progress

First add Gross Factory Cost and Opening Stock of Work-in-Progress: 3,93,000 + 79,000 = 4,72,000

Then subtract the Closing Stock of Work-in-Progress figure from Step 5: 4,72,000 – 24,562.50 = **₹4,47,437.50**

Step 7 — Add the Administrative Overheads to get Cost of Production.

First, total the Administrative Overheads: Administrative Overheads = Director's Fees + Office Salaries + Audit Fees
Administrative Overheads = 13,000 + 7,500 + 2,500 = **₹23,000**

Then apply the formula from Section 3:

Cost of Production = Factory Cost + Administrative Overheads

Cost of Production = 4,47,437.50 + 23,000 = **₹4,70,437.50**

Step 8 — Attempt to move to Cost of Goods Sold, and hit a data problem.

The next line in the template requires the Opening Stock of Finished Goods and the Closing Stock of Finished Goods. We are given the Opening Stock of Finished Goods both in units (3,500 units) and in rupees (₹35,000) — so that side is not a problem. But the Closing Stock of Finished Goods is not given in any form at all — not in units, and not in rupees. So, following the same logic as before, we turn to the Section 3 formula that derives Closing Stock of Finished Goods in units from the other unit figures we have:

Closing Stock of Finished Goods, in units = Opening Stock of Finished Goods in units + Units Produced – Units Sold

Substituting the numbers given:

Closing Stock of Finished Goods in units = $3,500 + 32,000 - 38,000$

Breaking this into two steps: $3,500 + 32,000 = 35,500$ (this is the total number of units that were available to sell during the period — the ones already sitting in stock at the start, plus everything freshly produced). Then, $35,500 - 38,000 = -2,500$ **units**

Why this result is a genuine problem, explained fully: a Closing Stock figure represents however many units are physically left over and unsold at the end of the period. It is mathematically and physically impossible for this number to be negative — you cannot have "negative two thousand five hundred units" sitting in a warehouse. A negative result here specifically means that the business is shown to have sold more units (38,000) than it ever had available to sell in total (35,500, being the 3,500 already in stock at the start plus the 32,000 freshly produced during the period). The shortfall is exactly 2,500 units — that is, the question as transcribed is claiming 2,500 more units were sold than could possibly have existed.

This tells us, with certainty, that one of the three input numbers used in this formula has been transcribed incorrectly somewhere between the original textbook or question paper and the version you photographed — it could be the Opening Stock of Finished Goods in units (perhaps it should be a larger number), the Units Produced (perhaps it should be higher), or the Units Sold (perhaps it should be lower). Because your uploaded image shows the exact same three figures (32,000 produced, 38,000 sold, 3,500 opening) as the version already checked, this is not something that happened during our conversation — the inconsistency is present in the source material itself, and the only way to resolve it is to check the original textbook, question bank, or ask your professor which of the three figures is correct.

Step 9 — What remains once the correct figure is confirmed (kept ready for you to finish).

Once you have the corrected number of units for Closing Stock of Finished Goods, here is exactly how to finish the question, following the same template and the same conversion logic used throughout this guide:

First, convert the corrected Closing Stock of Finished Goods from units into rupees, using the Section 3 formula:

Closing Stock of Finished Goods in rupees = (Cost of Production ÷ Units Produced) × Closing Stock of Finished Goods in units

Note that this uses Cost of Production (₹4,70,437.50) and Units Produced (32,000) — not Gross Factory Cost — because by this stage in the template we have already moved one stage further along the chain, and the "cost per unit" being used should reflect the fuller cost including Administrative Overheads.

Second, calculate Cost of Goods Sold using the Section 3 formula:

Cost of Goods Sold = Cost of Production + Opening Stock of Finished Goods – Closing Stock of Finished Goods

Here, use the ₹35,000 figure already given for Opening Stock of Finished Goods (there is no need to convert this one — it was given to us directly in rupees, not just in units).

Third, calculate Cost of Sales:

Cost of Sales = Cost of Goods Sold + Selling and Distribution Overheads

No Selling and Distribution Overhead items (such as advertising, sales commission, or carriage outwards) were listed anywhere in this extract of the question — so check whether the original question paper includes any such items elsewhere that were not captured in your photograph. If truly none exist, then Cost of Sales simply equals Cost of Goods Sold.

Fourth, calculate Sales, which in this question can be found independently of everything above, using only the units actually sold and the selling rate:

Sales = Units Sold × Selling Rate per unit
Sales = 38,000 × 15 = **₹5,70,000**

Fifth and finally, calculate Profit using the Section 3 formula, rearranged:

Since Sales = Cost of Sales + Profit, rearranging gives:

Profit = Sales – Cost of Sales
Profit = 5,70,000 – Cost of Sales (once Cost of Sales is known from Step 3 above)

Summary of what has been solved so far in Question 2, pending the data correction:

Stage	Result
Gross Factory Cost	₹3,93,000.00
Factory Cost	₹4,47,437.50
Cost of Production	₹4,70,437.50
Cost of Goods Sold	cannot be finalized — depends on the corrected Closing Stock of Finished Goods
Cost of Sales	depends on Cost of Goods Sold above
Sales (independently calculable)	₹5,70,000.00
Profit	depends on Cost of Sales above